

Proposal: Dynamically Controlled Travelling Plasma Grating for Relativistic High-Harmonic Generation

Proposal ID:

eli80069

Brief summary:

Sub-cycle plasma dynamics fundamentally govern surface high-harmonic generation (SHHG). While temporal synthesis of femtosecond pulses using collinear two-color fields enables control over sub-cycle electron motion, non-collinear interference introduces a spatiotemporal control. In the second case, the fundamental and second-harmonic pulses generate a transient, traveling grating-like structure on the plasma surface. This work proposes, for the first time, imprinting a traveling transient plasma grating formed by interfering high-power, noncollinear, two-color pulses onto a free-flowing, ultrathin, replenishable liquid-sheet target. A third few-cycle, relativistic-intensity pulse, derived from the same laser, interacts with the traveling transient plasma grating, resulting in SHHG via the Relativistic Oscillating Mirror (ROM) mechanism. As the traveling transient plasma grating surface evolves at sub-cycle time scale, it directly impacts the characteristics of the generated harmonic beam. Therefore, this approach will provide a unique diagnostic handle to probe the sub-cycle dynamics of high-density electron "nano-spikes" that generate harmonics via ROM. Since the precise control over the spatio-temporal evolution of the dynamic plasma grating is achieved through the relative phase between the two colors, this will allow the mapping of relativistic surface dynamics with unprecedented spatial and temporal resolution.

Proposal Team**Principal Investigator:**

Sudipta Mondal

Anubal Fusion

Co-proposer:

Prashant Kumar Singh, TIFR Centre for Interdisciplinary Sciences

Subhendu KAHALY, ELI-HU Research and Development Non-Profit

Indranuj Dey, ELI-ALPS

Ravishankar Sugumar, Anubal Fusion

Moniruzzaman Shaikh, Anubal Fusion

Abhisek Sinha, Anubal Fusion

Shivam Kumar Mishra, Anubal Fusion

Imran Khan, Anubal Fusion

Mojtaba Shirozhan, ELI-HU Research and Development Non-Profit

Proposal Identification

Select instrument

SYLOS3-SYLOS SHHG

Would you like to choose a secondary (alternative and/or additional) instrument? No

Acronym

LIQGRATHHG

Basic Information

Peer review panel Relativistic and ultrarelativistic Interactions

Please indicate here if this is a resubmission or a continuation of a previous proposal New proposal

Previous ID (ELIUPMx-XX or eli5xxxx)

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Rationale

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Societal challenge Energy

Unavailable weeks

ISO week number: 2026: W49-W50; 2027: W1-W2; W5

Have you contacted the instrument team? Yes

Name of your ELI contact

Subhendu Kahaly, Indranuj DEY, Mojtaba Shirozhan

Do you need support in data analysis No

Estimated total amount of data generated by the experiment between 1 TB and 10 TB

Estimated total number of data files generated by the experiment 100000

Experimental Team

Dr. Sudipta MONDAL (Affiliation: Anubal Fusion; Expertise: Team lead, Relativistic Laser-Matter Interaction, Surface High Harmonic Generation, XUV Diagnostics), (email: msudiptaphy@gmail.com); Dr. Moniruzzaman Shaikh, (Affiliation: Anubal Fusion; Expertise: Relativistic Laser-Matter Interaction and XUV diagnostics), (email: moniruzzaman.shaikh@anubalfusion.com) Dr. Ravishankar Sugumar, (Anubal Fusion; Expertise: Relativistic Laser-Matter Interaction, Data Analysis, X-ray and Electron Diagnostics), (email: ravishankar.sugumar@anubalfusion.com) Dr. Abhisek Sinha, (Anubal Fusion; Relativistic Laser-Matter Interaction, Data Analysis and XUV diagnostics), (email: abhisek.sinha@anubalfusion.com) Dr. Prashant Kumar Singh, (Tata Institute of Fundamental Research; Expertise: Team lead, Relativistic Laser-Matter Interaction, Surface High Harmonic Generation, XUV and Electron Diagnostics), (email: pks@tifrh.res.in) Dr. Subhendu Kahaly, (ELI-HU Research and Development Non-Profit, ELI ERIC; Expertise: Team lead, Relativistic Laser-Matter Interaction, Surface High Harmonic Generation, XUV and Electron Diagnostics), (email: Subhendu.Kahaly@eli-alps.hu) Dr. Indranuj Dey, (ELI-ALPS; Expertise: Team lead, Relativistic Laser-Matter Interaction, Surface High Harmonic Generation, XUV and Electron Diagnostics), (email: Indranuj.Dey@eli-alps.hu) Dr. Mojtaba Shirozhan, (ELI-ALPS; Expertise: Relativistic Laser-Matter Interaction, Surface High Harmonic Generation, PIC Simulation, XUV Diagnostics), (email: Mojtaba.Shirozhan@eli-alps.hu)

Scientific Objectives

Objectives of the proposed experiment

We propose to utilize the SYLOS 3 - SHHG beamline at ELI-ALPS to investigate surface high-harmonic generation (SHHG) from relativistic plasma mirrors formed on free-flowing liquid sheet targets. The primary objective is to implement a spatiotemporal gating by interfering with two non-collinear laser pulses of distinct frequencies: the fundamental (ω) and its second harmonic (2ω). By operating at intensities of $\sim 10^{18}$ W/cm², we aim to move beyond traditional collinear two-color synthesis into a regime of spatiotemporal wavefront engineering.

In a non-collinear geometry, the interference of ω and 2ω fields generates a transient intensity modulation characterized by a "traveling grating" structure. Unlike static gratings, this interference pattern evolves dynamically across the target surface. The phase velocity of these intensity fringes, v , is governed by the projection mismatch of the wave vectors onto the target surface. For a symmetric incidence angle θ relative to the surface normal, the transverse velocity of the grating is defined by [1]

$$v = c / 2\sin(2\theta)$$

This superluminal phase velocity allows the laser-plasma interaction to be "swept" across the liquid surface on a sub-cycle timescale of the driver laser.

The travelling grating effectively triggers ROM-based harmonic emission at spatially and temporally shifted intervals across the focal spot. This creates a phased-array effect, in which the local emission phase of the harmonics is coupled to the grating velocity and to the relative phase $\Delta\phi$ between the two colors.

By precisely tuning the carrier-envelope phase (CEP) and the inter-pulse delay, we can modulate the spatial structure of the plasma surface and the resulting harmonic beam characteristics. This setup provides a high-resolution diagnostic handle to extract the dynamics of high-density electron "nano-spikes" responsible for ROM harmonics, thereby allowing us to map the relativistic trajectories of the plasma surface with unprecedented precision.

Due to the thin (sub-micrometer) nature of the liquid sheet target, harmonic emission is also expected in the transmission direction. The XUV emission will be analyzed with a flat-field grating spectrometer, both in the specular reflection and transmission direction along with spatio-spectral characterization of the emitted electrons using a magnetic electron spectrometer and a LANEX screen.

Scientific background and rationale

Surface high-order harmonic generation (SHHG) occurs during the interaction of high-intensity, short-pulse lasers with overdense plasma surfaces, providing a robust mechanism for producing attosecond bursts of extreme ultraviolet (XUV) radiation [2,3]. When an intense, ultrashort laser pulse interacts with a solid-density plasma, the strong laser field drives ultrafast electron "nano-spikes" that oscillate at sub-cycle timescales [4]. These relativistic plasma oscillations act as a dynamic mirror; as the laser pulse reflects off this oscillating boundary, its phase is periodically modulated. In the frequency domain, this temporal phase distortion manifests as a coherent spectrum of high-order harmonics [2].

Therefore characteristics of the surface high harmonics can greatly be optimized by tuning the spatio-temporal dynamics of the electron "nano-spikes" [5-9]. In the spatial domain, sinusoidal plasma grating generated by two interfering high power laser pulses on an overdense plasma surface is employed to control and extract the source parameters [2,10]. Whereas temporal synthesis of the driving field using two-color laser pulses is employed to precisely manipulate ultrafast electron "nano-spike" dynamics, thereby optimizing the emission of attosecond bursts [5-8]. Precise CEP tuning in a few-cycle laser pulse can manipulate the electron "nano-spike" emission with 100's attosecond time precision thereby controlling the attosecond bursts from surface plasma [9].

The synthesis of a two-color driver pulse is often achieved in a co-linear geometry by combining a fundamental laser with its second harmonic [8]. However, in this configuration only temporal domain manipulation of the driver field waveform can be achieved. On the other hand, spatial modulation on a plasma surface is commonly induced through the interference of two noncollinear laser pulses. The resulting plasma grating is essential for surface high-harmonic generation because it can reveal spatio-temporal details of attosecond bursts that were inaccessible [1].

In this work, for the first time, we propose interfering high-power laser pulses of different wavelengths (fundamental ω and its second harmonic 2ω) at an angle 2θ between them in a non-collinear geometry to form a traveling transient plasma grating on a replenishable ultrathin liquid sheet target at kHz repetition rate. A third high-power few-cycle laser pulse interacting with the traveling transient plasma grating target generates surface high-harmonics. Therefore, this configuration provides simultaneous access to the spatio-temporal dynamics of the surface high-harmonic generation process at sub-cycle time precision.

Rationale: The success of this experiment relies on high shot-to-shot stability, broad parameter variations (preplasma optimization, relative delays between the grating imprinting pulses and main pulse, intensities, and angle between the grating-imprinting pulses), and precise control over the CEP. The SYLOS3 - SYLOS SHHG beamline, driven by the state-of-the-art CEP-stabilized SYLO3 laser system, provides a unique environment for these measurements. Furthermore, the integration of a replenishable, sub-micron-thick liquid target [13, 14] enables surface high-harmonics generation at kHz repetition rate [13], ensuring the required high statistics. These unique capabilities are essential for the proposed experiments that were difficult to perform in the past. These experiments may provide new insights into the generation and spatio-temporal control of surface high harmonics. Additionally, the traveling transient plasma grating target will provide spatio-temporal control over "vacuum laser acceleration" of relativistic electrons in the interaction of a few-cycle relativistic intensity laser interacting with a plasma mirror [15].

Preliminary work already carried out

Commissioning and user experiments at the SYLOS-SHHG beamline have successfully utilized liquid sheet targets to generate XUV radiation using the SYLOS 3 lasers. The liquid plasma mirror target has been upgraded for continuous kHz-rate operation through a computer-controlled dual-pump system, facilitating seamless integration with SYLOS 3 [11, 12]. Furthermore, plasma gratings on solid targets have been demonstrated as a proven technique for SHHG [10].

The proposer team has extensively investigated two-color (ω - 2ω) plasma synthesis in a collinear configuration using 2D PIC simulations [8]. Relativistically intense laser pulses drive ultrafast push-pull electron dynamics at the plasma-vacuum interface, triggering the emission of coherent surface high harmonics. This bicolor pulse synthesis provides a vital mechanism for sub-cycle waveform shaping, allowing for precise temporal control over the driver field and the resulting push-pull plasma dynamics. Optimizing the relative delay between the two-color (ω - 2ω) pulses enhances the harmonic flux and suppresses multiple attosecond bursts, producing a single, intense attosecond pulse.

The implementation of two-color gratings in non-collinear configuration has been successfully used for ptychographic measurements of SHHG, highlighting the significant potential of this approach [1].

Furthermore, preliminary 2D and 3D Particle-In-Cell (PIC) simulations have been carried out to gain insight into the ultrafast plasma dynamics and crosscheck the feasibility of the proposed experiments. The details of the PIC simulations are presented below.

Two Gaussian laser pulses (Gaussian both in time and space) are incident at an angle $\pm$ of 15° on a thin solid target (thickness $1\text{ }\mu\text{m}$) with electron density $n_0=100n_c$ and preplasma scale-length $0.08\text{ }\mu\text{m}$. In one case (Figure 2(a)), we use both lasers at fundamental frequencies ($\lambda = 0.8\text{ }\mu\text{m}$); in the other (Figure 2(d)), one pulse is at the fundamental frequency ($\lambda = 0.8\text{ }\mu\text{m}$), and the other is at the second-harmonic frequency ($\lambda = 0.4\text{ }\mu\text{m}$). The intensity (10^{16} W/cm^2) and focal spot ($8\text{ }\mu\text{m}$) are the same for both fundamental-fundamental and fundamental and second harmonic cases.

In the first case (fundamental and fundamental), uniform, equidistant interference patterns are generated on the target plane (Figure 2(b)), resulting in a transient sinusoidal plasma grating after interaction, as shown in Figure 2(c). This grating persists on a picosecond timescale and has unique applications in surface high-harmonic generation [10].

In the second case (fundamental and second harmonic), a traveling interference pattern with speed $v = c / 2\sin(2\theta) = c / 2\sin(30) = c$ is generated at the target plane (Figure 2(e)), which leads to the formation of a transient traveling plasma grating as shown in Figure 2(f). The generated grating structures are similar to the electron “nano-spikes” that result in the generation of surface high-harmonics in relativistic-intensity laser-solid interactions. Therefore, the transient traveling plasma grating can influence surface high-harmonic generation and reveal the sub-cycle dynamics of the electron “nano-spikes.”

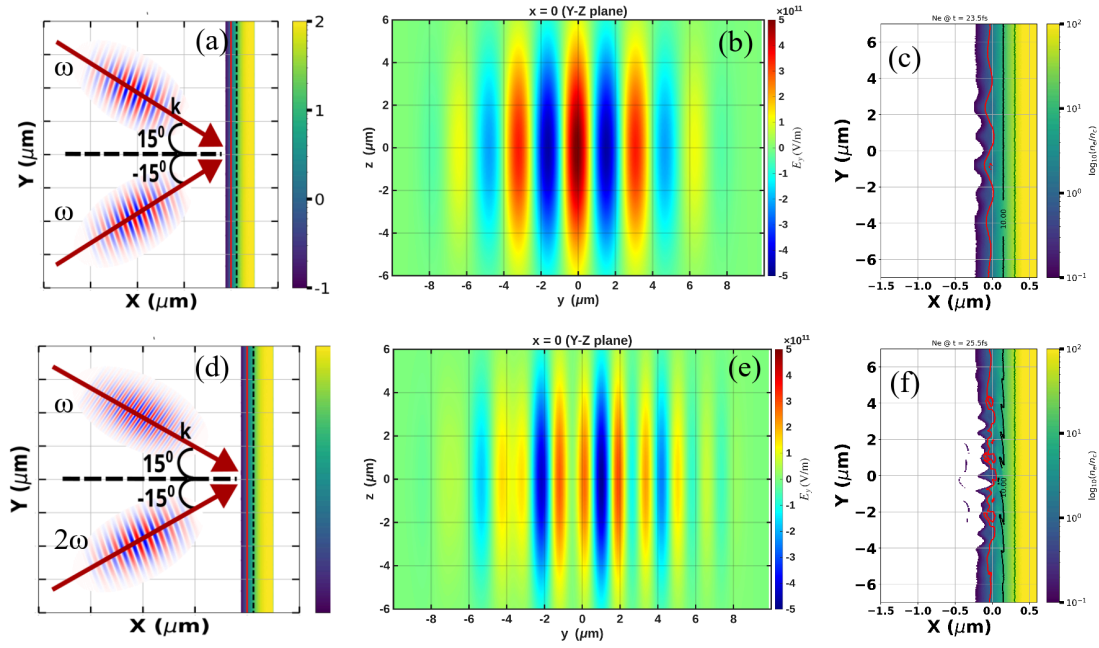


FIGURE 2: (a) 2D PIC setup for two-beam of same color (ω) grating formation. (b) Simulated E_y field at the target plane. (c) Electron density at 23.5 fs. (d) 2D PIC setup for two-color interference (ω and 2ω). (e) Simulated E_y field at the target plane showing a traveling interference pattern ($v = c / 2\sin(2\theta)$). (f) Electron density at 25.5 fs exhibiting "nano-spikes." Red curve: critical density (n_c) contour in (c) and (f). Densities below $0.1n_c$ are omitted.

PIC simulation results indicate the formation of a travelling interference pattern as well as travelling plasma grating on the surface during the laser pulse interaction with the target.

In this proposal, building on prior work in both experiments and simulations, we intend to test a unified framework to optimize surface high-harmonic generation from a traveling transient plasma grating at relativistic laser intensity by simultaneously varying the delay between the grating-forming pulses.

Relevant publications and references

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14. K. Osvay et al., Eur. Phys. J. Plus **139**, 574 (2024). <https://doi.org/10.1140/epjp/s13360-024-05338-1>
15. M. Thévenet, et al., Nature Phys **12**, 355 (2016). <https://doi.org/10.1038/nphys3597>

Technical details

Methodology and feasibility

Experimental Setup:

Experiments will be performed at the ELI-ALPS using the SYLOS-SHHG beamline, employing the SYLOS 3 laser (~ 8 fs, ~ 70 mJ on target) to drive high-order harmonic generation (HHG) from a relativistically oscillating plasma mirror (ROM). The laser pulse is focused by a 100 mm focal-length off-axis parabola (OAP) at a 55° angle of incidence on target, yielding a tight focal spot ($< 2.5 \mu\text{m}$ FWHM) [11,12].

Target and Interaction:

Two half-inch mirrors will be placed in front of one of the flat mirrors as shown in Figure-1 (a) in the interaction chamber to extract two prepulses of about 1-5 mJ of energy. One of the prepulses would be converted to a second harmonic using a thin Type-I BBO crystal as shown in Figure-1 (a). A set of relevant beam control e.g. polarization, energy, attosecond delay etc will be installed on one of the prepulse arms. The attosecond delay will be incorporated by precisely tilting a thin calcium fluoride (CaF_2) plate in the second-harmonic beam. Figure-1 (b) shows the current liquid sheet target setup installed in the interaction chamber. The diagnostic layout and the traveling grating are shown in Figure. 1(c) and Figure. 1(d), respectively.

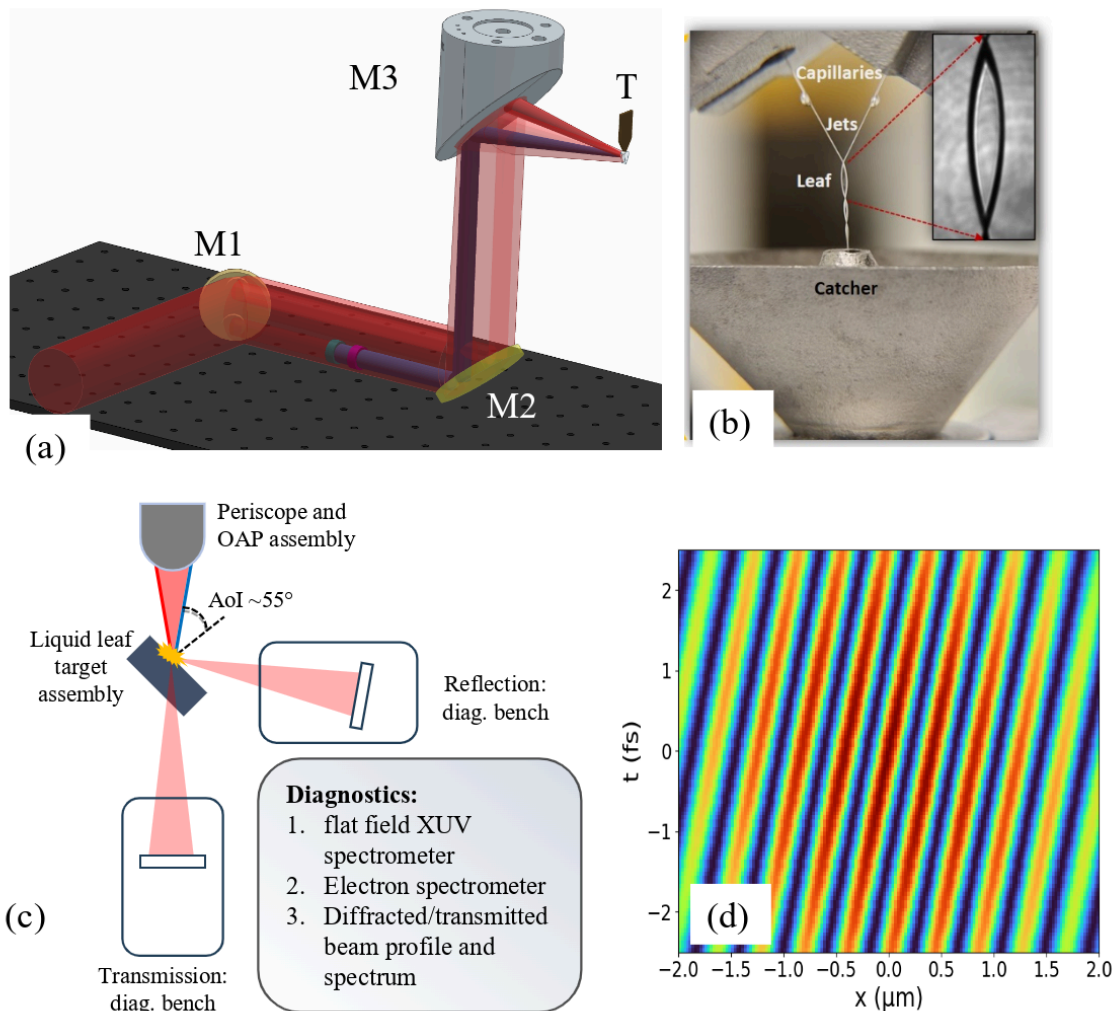


FIGURE 1: (a) Schematic of the interaction configuration. Interfering beams are extracted by placing two half-inch step-mirrors in front of M1. A thin Type-I BBO generates 2ω , with the fundamental filtered by a dichroic mirror. A motorized CaF_2 plate provides attosecond 2ω delay. An OAP mirror (M3) is used to focus all three beams on the liquid sheet target (T). (b) A photo of the liquid sheet target in the beamline. (c) Interaction and diagnostic layout. (d) Transient optical

grating resulting from the superposition of ω and 2ω beams at focus, as a function of time (t) and position along the modulation direction (x) of the object in the surface plane.

Sub-micron thick liquid sheet targets (already integrated in the beamline) will be used as the target. The experiments will be performed in the following steps.

1. A prepulse (already installed in the setup) will be used to generate optimized preplasma for SHHG.
2. Two color pulses interacting with the target at an angle θ with respect to the target normal will form a travelling grating on top of the preplasma.
3. A few-cycle main pulse will interact with the travelling grating and generate high-harmonics. Due to the ultrathin nature of the target, high-harmonics can be detected in specular reflection as well as in the transmission direction.

In this beamtime, we are aiming to achieve the following

1. Transient travelling plasma grating on ultrathin replenishable liquid sheet target. This will demonstrate the technical feasibility of structured or shaped plasma optics for high-repetition-rate laser plasma applications.
2. The control of plasma electron dynamics with sub-cycle temporal precision.
3. Since a small change in the surface plasma will directly affect the SHHG. Therefore SHHG can be used as a diagnostic for the transient traveling plasma grating.
4. Direct control of the sub-cycle electron dynamics on the target surface and consequently vacuum relativistic electron acceleration by the laser field from the target surface.

Diagnostics:

The experiments will utilize a set of complementary diagnostics and monitoring tools. XUV harmonics emitted in both reflection and transmission geometries will be analyzed using flat-field grating spectrometers coupled to MCP-phosphor assemblies. Simultaneously, two magnetic electron spectrometers with LANEX screens will record the electron energy distribution ejected from the target. These measurements will provide critical insight into the interaction dynamics, enabling correlation between electron acceleration and harmonic generation. To diagnose spatial modulations on the plasma surface, the diffracted main pulse will be monitored.

Feasibility:

Built on established infrastructure and validated techniques, this project is highly technically feasible. At its core, the SYLOS 3 laser at ELI-ALPS delivers the precise pulse energy and few-cycle duration essential for plasma mirror High-order Harmonic Generation (HHG). By employing liquid sheet targets, proven to operate at high repetition rates, we maintain optimal vacuum conditions and minimize XUV absorption, both are critical for detecting weak harmonics. This integration of state-of-the-art laser technology, non-collinear two-color interference, and a dual photon-electron diagnostic plan provides a versatile framework for attosecond science experiments.

Beamtime requirements**Preparatory time (1 week): with alignment diode laser**

One week of preparatory time is needed with alignment diode laser to install optics for the two additional interfering pulses and install all the diagnostics in their approximate location.

Alignment test and experiment (3 weeks): with SYLOS-3 laser

Low power main beam (1mJ-5mJ) to align through the beamline and set reference.

Focal spot imaging will need ~ nanojoule energy main beam at focal spot. The same imaging system will image the focal spot of the prepulse, main-pulse and the interference at the focal plane. Therefore, energy tunability in the main laser is highly required.

During data recording the following parameters of the lasers will be required.

Laser: SYLOS 3

Energy: Tunable; 20% to 100% of the maximum possible energy (~ 120 mJ on laser side).

Pulse duration: 8-11 fs for the experiments. Precise dispersion optimization may be needed during SHHG optimization

Polarization: P-polarized on target

Repetition rate: 100 Hz and 1000 Hz.

CEP tagging and stabilization.

Collinear prepulse:

Energy: ~ 1mJ

Pulse duration: < 50 fs

Prepulse delay: ± 150 ps

Interfering pulses:

Fundamental (ω)-Fundamental (ω) as well as Fundamental (ω)-Second harmonic (2ω) interfering pulse generation and beam control system will be developed in consultation with the beamline scientists by modifying one of the suitable mirrors in the beamline as shown in Figure 1(a). This will be developed together by the user and beamline scientist to ensure full adaptability and functionality.

Beamtime request (days) 15

Laser/ Beam requirements**Laser:** SYLOS 3

Low power main beam (1mJ-5mJ) to align through the beamline and set reference.

Focal spot imaging will need ~ nanojoule energy main beam at focal spot. The same imaging system will image the focal spot of the prepulse, main-pulse and the interference at the focal plane. Therefore, energy tunability in the main laser is highly required.

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Endstation requirements

Optics for beam transport and focusing, as well as a collinear prepulse for preplasma optimization, are already present in the beamline. Additionally, two non-collinear prepulses need to be generated, with precise control over polarization, energy, and delay. It will be developed in consultation with the beamline scientists before the experiments. One of these prepulses needs to be at the fundamental frequency, and another at the second harmonic frequency. Two XUV spectrometers, electron spectrometers, focal spot imaging, and preplasma characterization systems are already installed in the beamline.

Facility requirements

Mechanical design support, electrical workshop support, software development and machinework support as it is required during the technical discussion as well as during the beamtime.

Standard cables, BNC cables, high voltage cables, power extension board, gloves.

Standard tools, optics and optics cleaning accessories required in a modern optics laboratory, cleanroom.

Liquid nitrogen for cold traps.

Experimental Plan

Assuming the beamline is well aligned with NF-FF references, D-Scan, CEP setup and focal-spot is aligned in air; liquid leaf target is installed and tested, XUV spectrometers are installed at approximate location, the experiments will be carried out in the following steps:

Week 1:

- Setting up two-color interfering pulses in the interaction chamber (VC-3).
- Spatial and temporal matching of the main-pulse, pre-pulse and two-color interfering beams in air.
- Focal spot optimization in vacuum and shot to shot stability measurement.
- Chirp-scan followed by D-Scan to optimize the pulse-width in vacuum.

Week 2:

- Spatio-temporal matching of the main-pulse, pre-pulse, and two-color interfering in vacuum.
- Alignment of the XUV and electron spectrometers in the reflection and transmission directions.
- Light shielding of the diagnostics, while ensuring flexibility of movement wherever required.
- Checking the generation of HHG from the plain plasma (liquid sheet) target.
- Preplasma optimization for the efficient High-harmonics generation.
- Spatial Domain Interferometric (SDI) measurement for preplasma scale-length

Week 3:

- Generation of HHG from the transient plasma grating target using the same color (ω - ω).
- Generation of HHG from the transient Travelling plasma grating target using the two-color (ω - 2ω).
- Systematic recording of high-harmonics spectrum and spatial profile with and without travelling plasma grating and CEP scan.
- Systematic recording of electron spectrum and spatial profile with and without plasma grating and CEP scan.

Diagnostics

1. Two XUV flat-field grating spectrometers with MCP-Phosphor assembly and subsequent imaging system for reflection and transmission directions.
2. Two lanex based magnetic spectrometers and associated imaging system for electron spectrum for reflection and transmission directions.
3. Reflected main pulse in the specular reflection direction on spectralon with imaging camera and lenses.
4. SDI imaging for pre-plasma scale-length estimation.
5. CEP tagging (+ feedback) setup.
6. Laser pulse-width and contrast measurement.
7. NF-FF diagnostics and focal spot imaging system.

Additional technology, equipment, and service requirements provided by ELI

Mechanical design, electrical workshop support, software development and machine workshop support as required during the technical discussion and during beamtime.

Standard, cables, BNC cables, high-voltage cables, power extensions *etc.*

Gloves, standard tools needed to work in a modern optical laboratory.

Liquid nitrogen for cold traps.

Do you plan to bring and install your own equipment for the experiment? No

Samples, Targets and Materials

Will you provide samples, targets and materials? No

Do you request samples, targets and material from ELI? Yes

Samples, targets, and materials (precursor and / or target specifications) requested from ELI

1 of 2

Material

Ethylene Glycol

Type

Standard

In situ preparation procedure (if applicable)

Left blank

Estimated amount

2 liters

Handling requirements

Handling using gloves and eye protection

All associated hazards *Left blank*

2 of 2

Material

Type-I BBO crystal for SHG

Type

Optical crystal

In situ preparation procedure (if applicable)

20 micrometer thick 12 mm diameter Type-I BBO crystal for second harmonic generation

Estimated amount

1

Handling requirements

None

All associated hazards *Left blank*

Compliance

I have read and accept the Terms and Conditions for Access. Yes

I have read and accept the GDPR Information Notice. Yes

In my capacity as PI, I informed team members participating in this Proposal about the Access Terms and Conditions and GDPR Information, acknowledging that their acceptance of these documents is an admission condition to ELI Experiments. Yes

I have completed the scientific and technical content of the Proposal to the best of my knowledge. I understand that omitting or inaccurately providing important technical information or associated hazards could result in refusal of the proposal. Yes